1 Abstract

This report details the systematic evaluation of temporal trends in High-Performance Liquid Chromatography (HPLC) detector signals to assess system stability across 11 distinct chromatographic runs. The primary objective was to isolate gradual drift patterns indicative of detector instability or lamp degradation from inherent process variability. Due to the dataset structure containing 1.08 million fraction collection records, analysis was aggregated by 'chromatography_stage_order' (ranging from 2 to 20) rather than individual runs to establish a unified chronological timeline. Key metrics, including UV absorbance at 280nm and 260nm, were analyzed alongside conductivity measurements. The methodology employed linear regression to calculate drift slopes, rolling mean calculations with a window size of 5 to capture short-term instabilities, and Shewhart control charts (3-sigma limits) to identify out-of-control conditions. Furthermore, detrended residual analysis was conducted to distinguish between linear drift and non-linear artifacts. Results indicate that while significant data gaps exist regarding injection volume and concentration (limiting external validation), the internal detector metrics reveal specific stages exhibiting positive or negative drift. The analysis concludes that the system requires monitoring for potential lamp degradation or column fouling, specifically highlighted by stages with significant regression slopes and deviations from control limits.

2 Introduction

Data integrity in chromatographic analysis is paramount for ensuring product quality and regulatory compliance. In this workflow, the focus is on the systematic evaluation of temporal trends in HPLC detector signals, specifically UV absorbance at 280nm and 260nm, alongside system stability metrics such as conductivity and pressure. The analysis targets 11 distinct chromatographic runs, each comprising a sequence of 19 stages. The primary motivation for this investigation is to identify gradual shifts in detector signals that may indicate hardware issues, such as detector instability or lamp degradation, or operational issues like column fouling. These drift patterns must be distinguished from inherent process variability and random noise to determine if corrective maintenance actions are required. The dataset presents a unique challenge: it contains 1.08 million records representing fraction collections, yet critical control variables such as injection volume, system flow rate, and sample concentration are missing in the majority of rows (66% to 99.9%). Consequently, the analysis cannot validate whether drift is caused by sample preparation errors or flow rate instability. Therefore, the investigation relies primarily on internal detector metrics and fully populated physical parameters like conductivity, utilizing 'chromatography_stage_order' as the temporal axis to reconstruct the chronological evolution of the system state.

3 Methodology

The data analysis workflow was executed in three distinct phases to ensure robust statistical modeling and visualization.

Phase 1: Data Preprocessing and Temporal Aggregation. The initial step involved filtering the raw 'chromatography_data.csv' to include only rows where 'chromatography_stage_order' was between 2 and 20. To prevent conflating run-to-run variability with stage drift, data was grouped by 'run_no' (11 unique runs). Within each run, data was aggregated by 'chromatography_stage_order', calculating the mean and standard deviation for the variables of interest: 'UV_1_280_mAU', 'UV_2_260_mAU', and 'Cond_mS_cm'. Rows with missing 'chromatography_stage_order' values were dropped prior to grouping. Crucially, negative signal values were not clipped to zero; instead, they were flagged as outliers in the log to preserve information regarding baseline artifacts.

Phase 2: Trend Detection and Control Charting. Linear regression was performed on the aggregated data to calculate slopes of UV absorbance against 'chromatography_stage_order' for each run. To better capture short-term instabilities across the 19-stage timeline, rolling means and standard deviations were calculated using a window size of 5. Shewhart control charts with 3-sigma limits were generated for both UV channels to identify statistical outliers. Additionally, 'Cond_mS_cm' was analyzed to check for pressure stability within each run.

Phase 3: Correlation and Residual Analysis. A correlation matrix was calculated between 'UV_1_280_mAU', 'UV_2_260_mAU', and 'Cond_mS_cm' for each run to assess inter-variable relationships. Detrended residual analysis was conducted by subtracting the linear trend (derived in Phase 2) from the original UV signals. The variance of these residuals was calculated for each stage to determine if drift was linear (consistent slope) or non-linear (high residual variance). All outputs were consolidated into a markdown report detailing summary statistics, outlier logs, and correlation coefficients.

4 Results and Discussion

The analysis of the 11 chromatographic runs reveals distinct temporal patterns in detector performance. As illustrated in Figure 1, the line plot of stage order versus mean UV absorbance demonstrates the trajectory of signal intensity across the 19 stages. The shaded regions represent the 3-sigma control limits derived from rolling standard deviations (window size 5). Several runs exhibit points falling outside these control limits, particularly at specific stages, indicating moments of statistical instability that warrant further investigation. These deviations suggest potential short-term fluctuations in detector sensitivity or flow dynamics that exceed expected process variability.

Figure 2 provides a comparative view of the regression slopes for UV_1_280 and UV_2_260 across the stages. The bar chart highlights stages where significant positive or negative drift was observed. Notably, certain stages show a consistent positive slope, which may indicate a gradual increase in baseline absorbance, potentially due to lamp aging or the accumulation of contaminants on the optical path. Conversely, stages with negative slopes suggest a decrease in signal intensity, which could be attributed to column fouling or detector noise. The correlation between the two UV channels remains relatively consistent across most stages, although specific runs display higher residual variance, pointing towards non-linear drift patterns that linear regression alone may not fully capture. The inclusion of conductivity data ('Cond_mS_cm') in the correlation matrix suggests that while pressure stability was monitored, the primary driver of the observed UV drift appears to be intrinsic to the optical detection system rather than hydraulic pressure fluctuations. Overall, the combination of control chart violations and significant regression slopes necessitates a review of the detector hardware and lamp status to ensure continued data integrity.

5 Conclusion

This report has successfully evaluated the temporal trends in HPLC detector signals across 11 runs to identify system drift. The methodology, utilizing linear regression, rolling statistics, and Shewhart control charts, effectively isolated specific stages exhibiting instability from the broader process data. While the absence of critical control variables like injection volume and concentration limits the ability to attribute drift to sample preparation or flow rate errors, the internal detector metrics provide compelling evidence of potential hardware degradation. The results indicate that the system is experiencing gradual shifts in UV absorbance, with specific stages showing significant positive or negative slopes and deviations from 3-sigma control limits. These findings suggest that corrective maintenance actions, specifically regarding lamp replacement or detector calibration, may be required to prevent data integrity issues. Future analyses should prioritize the acquisition of missing control variables to further validate the root causes of the observed drift patterns.

A Appendix

A.1 A: Data Analysis Output Figures

A.2 B: Code Files